And even without gravitational waves, the amount of serious science in the movie was enormous. So when Chris sought ways to simplify *Interstellar*'s rich panoply of science, gravitational waves were a natural candidate for the ax. He jettisoned them.

For me, personally, Chris's decision was painful. I cofounded the LIGO Project in 1983 (together with Rainer Weiss at MIT and Ronald Drever at Caltech). I formulated LIGO's scientific vision, and I spent two decades working hard to help make it a reality. And LIGO today is nearing maturity, with the first detection of gravitational waves expected in this decade.

But Chris's reasons to jettison gravitational waves were compelling, so I didn't utter even a whisper of protest.

Gravitational Waves and Their Detectors

I indulge myself and tell you a bit more about gravitational waves before moving back to *Interstellar.*

Figure 16.6 is an artist's conception of some tendex lines emerging from two black holes that orbit each other counterclockwise, and collide. Recall that tendex lines produce tidal gravity (Chapter 4). The lines emerging from the holes' ends stretch everything they encounter, including the artist's friend, whom she has placed there. The lines emerging from the collision region squeeze everything they encounter. As the holes orbit around each other, they drag their tendex lines around, splaying outward and backward, like water from a whirling sprinkler.